# Trigonometry: Modelling in Radians

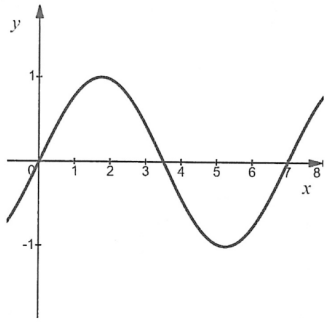
Questions 1–10

**Question 1.** State the period of each of the following functions:

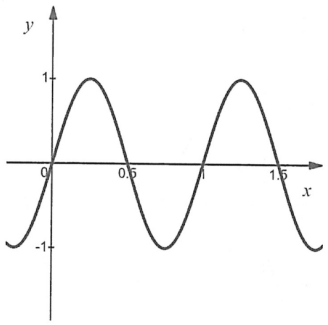
a) $f(x) = \sin(x)$
b) $f(x) = \cos(4\pi x)$
c) $f(x) = \tan(\pi x)$
d) $f(x) = \sin(\frac{\pi}{2}x)$
e) $f(x) = \cos(\frac{\pi}{4}x)$
f) $f(x) = \tan(\frac{\pi}{2}x - \frac{\pi}{4})$
g) $f(x) = \sin(\frac{2\pi}{52}x)$
h) $f(x) = \cos(\frac{2\pi}{7}x + \frac{\pi}{18})$
i) $f(x) = \sin(\frac{2\pi}{365}(x - 30))$
j) $f(x) = \cos(\frac{\pi}{26}(x + 7))$

**Question 2.** In each of the following cases, the curve $y = \sin x$ has been stretched parallel to the $x$-axis. State the equation of each curve.

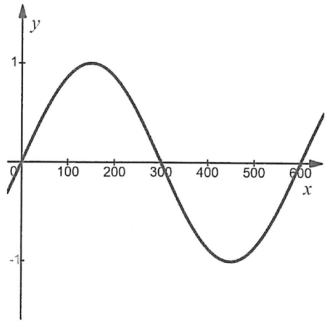
**a)**



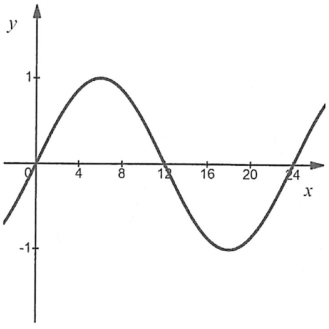

**b)**




**c)**

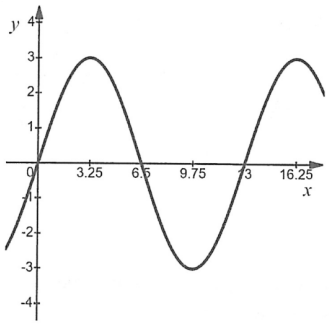



**d)**

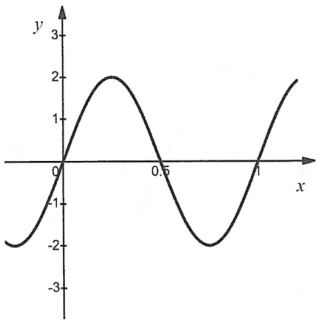


**Question 3.** All the following graphs are transformations of $y = \sin x$
State the equation of each:

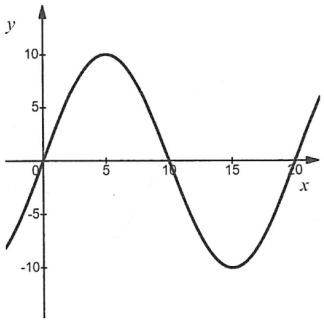
**a)**



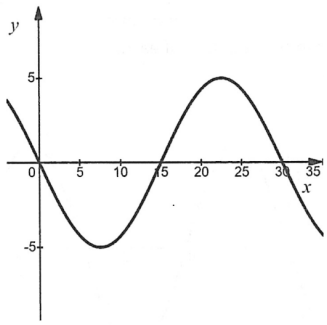

**b)**




**c)**




**d)**




**Question 4.** State the amplitude of each of the following periodic functions:

a) $y = \sin(3\pi x)$
b) $y = 4 + \cos(\pi x + 0.1)$
c) $y = 55 + 12\sin(\pi x)$
d) $y = 1 - 2\sin(x + \frac{\pi}{2})$

**Question 5.** For each of the following functions state the minimum and maximum of the function:

a) $y = \sin(2\pi x)$

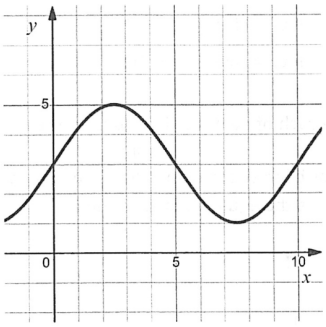
b) $y = 3 + \cos(\pi x - \frac{\pi}{2})$

c) $y = 14 + 6\sin(30\pi x)$

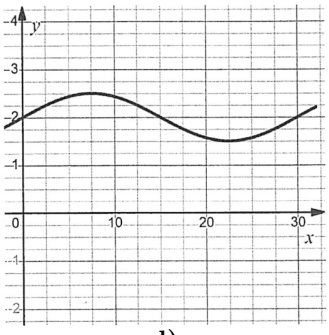
d) $y = 2 - 7\cos(\frac{\pi}{2}x + 0.2)$

**Question 6.** All the following graphs are all the form $y = a + b\sin(cx)$
In each case, find the equation of the graph:

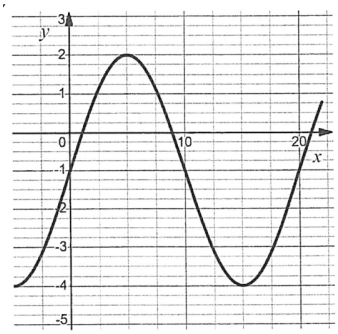
**a)**



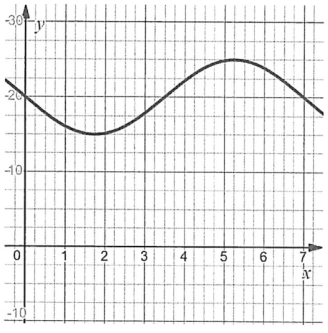

**b)**




**c)**




**d)**



**Question 7.** The river Hamble is tidal. On one day the depth of the Hamble oscillates between a maximum of 3.6 metres at 8:00 am and a minimum of 2.0 metres at 10:30 pm. Kaelen decides to model the depth of the river, $d$ metres by the equation

$$d = a + b\cos\left(\frac{2}{5}\pi t\right)$$

where $t$ is the number of hours since 8:00 am.

a) State the values of $a$ and $b$ that Kaelen should use.
b) State the period of this equation.
c) Calculate the total time, according to this model, that the Hamble is less than 2.5 metres in depth within the first 7 hours.

**Question 8.** The temperature, $T$°C of an average human can be modelled by the equation

$$T = 36.8 - 0.64\sin\left(\frac{\pi}{12}(h+2)\right)$$

where $h$ is the number of hours since midnight.

a) Find the maximum temperature of an average human.
b) Find the first time at which this maximum occurs after midnight.

The temperature, $T$°C of an average dog varies between 38.3°C and 39.2°C.

c) Find an equation to model a dog's temperature of the form $T = a - b\sin\left(\frac{\pi}{12}(h+2)\right)$

**Question 9. Prerequisite Knowledge for Q9: Graph Transformations (combinations)** The time of sunset, $t$ hours for the first day of the week $x$ for a year may be modelled by the equation

$$t = 18.5 + 2.4\sin\left(\frac{2\pi}{52}(x+26)\right)$$

for $0 \leq x \leq 52$

a) Describe the transformations which map

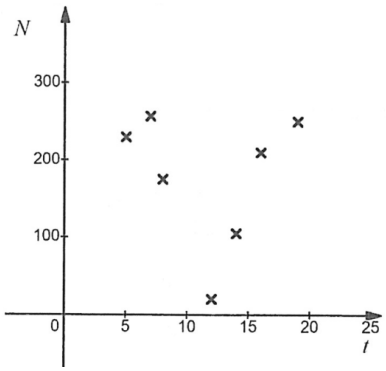
$$t = \sin\left(\frac{2\pi}{52}(x+26)\right)$$

onto

$$t = 18.5 + 2.4\sin\left(\frac{2\pi}{52}(x+26)\right)$$

b) Find the time of the sunset in the 14th week of the year.

**Question 10.** The number of passengers waiting on a platform for a train is recorded over the course of a day, and they are plotted on a graph. Passengers are only counted once as they embark on a train before the next count of passengers takes place.




It is believed that the data can be modelled with the equation

$$N = 140 + 120\sin\left(\frac{\pi}{6}(t-3)\right)$$

where $N$ is the number of passengers on the platform and $t$ is the number of hours after midnight. According to the model, find

a) the difference in the number of passengers between 5 am and 9 am

b) the times at which the total number of passengers on the platform is over 200 over the 24-hour period.

c) Explain why it may not be suitable to use this function to model the data.